New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

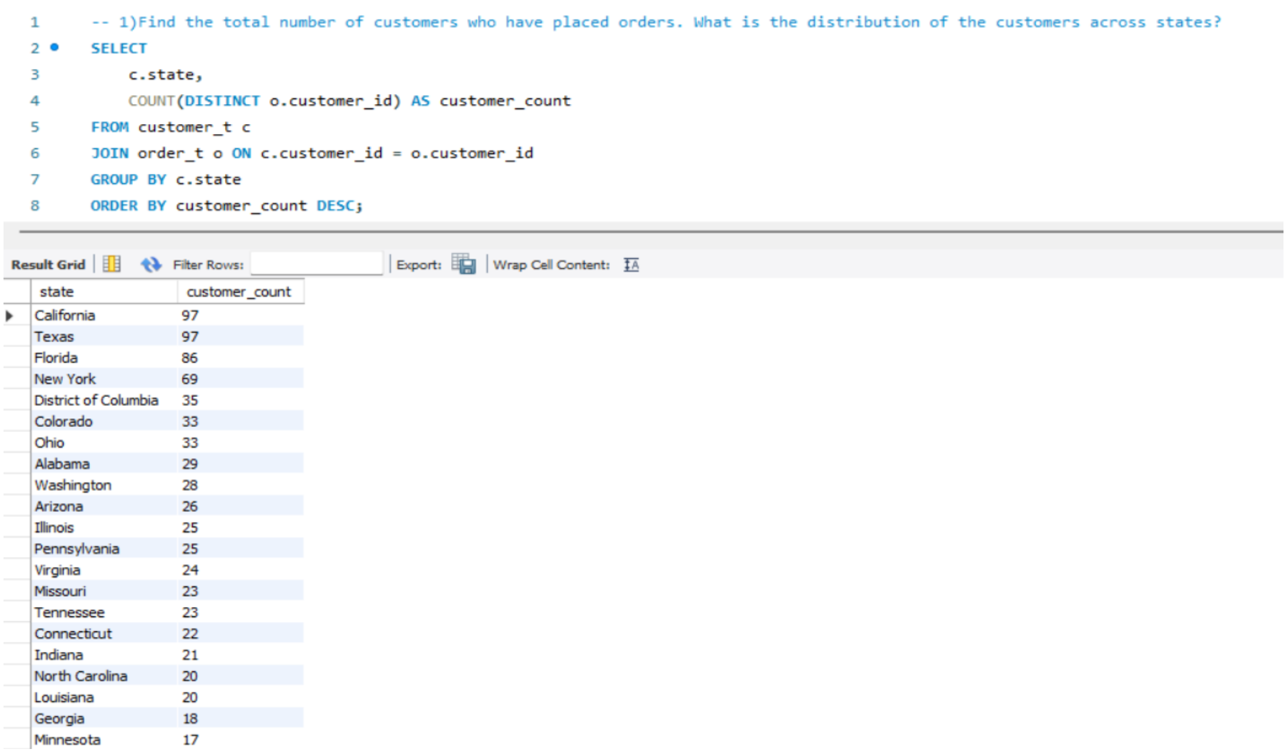
Business Questions

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```sql
SELECT
    c.state,
    COUNT(DISTINCT o.customer_id) AS customer_count
FROM customer_t c
JOIN order_t o ON c.customer_id = o.customer_id
GROUP BY c.state
ORDER BY customer_count DESC;
```

Output:



```sql
-- 1)Find the total number of customers who have placed orders. What is the distribution of the customers across states?
SELECT
    c.state,
    COUNT(DISTINCT o.customer_id) AS customer_count
FROM customer_t c
JOIN order_t o ON c.customer_id = o.customer_id
GROUP BY c.state
ORDER BY customer_count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

state	customer_count
California	97
Texas	97
Florida	86
New York	69
District of Columbia	35
Colorado	33
Ohio	33
Alabama	29
Washington	28
Arizona	26
Illinois	25
Pennsylvania	25
Virginia	24
Missouri	23
Tennessee	23
Connecticut	22
Indiana	21
North Carolina	20
Louisiana	20
Georgia	18
Minnesota	17

Observations and Insights:

- California (97) and Taxes (97) lead with the highest number of customers, has a significant share of total orders, these states are key markets focusing on these can bring further revenue.
- Florida (86) and New york (69) also have strong customer bases, indicating high market penetration in these states.
- District of Columbia (35), Colorado (33), and Ohio (33) show moderate customer engagement whereas Alabama (29) and Washington (28) maintained a consistent customer base.

Question 2: Which are the top 5 vehicle makers preferred by the customers ?

Solution Query:

```
SELECT vehicle_maker, count(vehicle_maker) as preferred_vehicle_maker
 FROM product_t group by vehicle_maker order by preferred_vehicle_maker desc limit 5;
```

Output:

```
-- 2)Write your query below
SELECT vehicle_maker,
COUNT(vehicle_maker) AS preferred_vehicle_maker FROM product_t
GROUP BY vehicle_maker
ORDER BY preferred_vehicle_maker DESC LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

vehicle_maker	preferred_vehicle_maker
Chevrolet	83
Ford	63
Toyota	52
Dodge	50
Pontiac	50

Observations and Insights:

- Chevrolet (83) is the most preferred vehicle maker, maintaining a significant lead over other brands. This shows strong customer loyalty and brand recognition for Chevrolet Vehicles.
- Ford (63) is in second rank and the gap suggests room for growth to catch up with Chevrolet.
- Toyota (52) holds the third position, showing the strong customer interest but falls behind Chevrolet and Ford.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
WITH vehicle_rank AS (
    SELECT
        c.state,
        p.vehicle_maker,
        COUNT(DISTINCT o.customer_id) AS No_of_purchase,
        RANK() OVER (PARTITION BY c.state ORDER BY COUNT(DISTINCT o.customer_id) DESC) AS ranks
    FROM order_t o
    JOIN customer_t c ON o.customer_id = c.customer_id
    JOIN product_t p ON o.product_id = p.product_id
    GROUP BY c.state, p.vehicle_maker
)
SELECT state, vehicle_maker, No_of_purchase
FROM vehicle_rank
WHERE ranks = 1
ORDER BY state;
```

Output:

state	vehicle_maker	No_of_purchase
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Cadillac	3
Arizona	Pontiac	3
Arkansas	Chevrolet	1
Arkansas	GMC	1
Arkansas	Mitsubishi	1
Arkansas	Pontiac	1
Arkansas	Suzuki	1
Arkansas	Volkswagen	1
California	Audi	6
California	Chevrolet	6
California	Dodge	6
California	Ford	6
California	Nissan	6
Colorado	Chevrolet	5
Connect...	Chevrolet	2
Connect...	Maserati	2
Connect...	Mercury	2
Connect...	Volvo	2
Delaware	Mitsubishi	2
District ...	Chevrolet	4
Florida	Toyota	7
Georgia	Toyota	3
Hawaii	Cadillac	1

Observations and Insights:

- 12 states show Chevrolet as the most preferred vehicle maker which includes Taxes (9), California (6) and Ohio (6).
- 7 states recognize ford as a leading choice such as Maryland (5), Virginia (5) and Michigan (3).
- California is highly competitive with 5 brands (Chevrolet, Dodge, Ford, Nissan and Audi) sharing equal preference.

Question 4: **Find the overall average rating given by the customers. What is the average rating in each quarter?**

Consider the following mapping for ratings: "Very Bad": 1, "Bad": 2, "Okay": 3, "Good": 4, "Very Good": 5

Solution Query:

```
WITH feedback_mapped AS (SELECT quarter_number, CASE WHEN customer_feedback = 'Very Bad' THEN 1
            WHEN customer_feedback = 'Bad' THEN 2
            WHEN customer_feedback = 'Okay' THEN 3
            WHEN customer_feedback = 'Good' THEN 4
            WHEN customer_feedback = 'Very Good' THEN 5 END AS feedback_score FROM order_t),
quarter_avg AS (SELECT quarter_number,
        AVG(feedback_score) AS avg_rating FROM feedback_mapped GROUP BY quarter_number),
overall_avg AS (SELECT AVG(feedback_score) AS overall_rating FROM feedback_mapped)
SELECT q.quarter_number, o.overall_rating, q.avg_rating FROM quarter_avg q
CROSS JOIN overall_avg o, ORDER BY q.quarter_number;
```

Output:

```
WITH feedback_mapped AS (SELECT quarter_number,
        CASE WHEN customer_feedback = 'Very Bad' THEN 1
            WHEN customer_feedback = 'Bad' THEN 2
            WHEN customer_feedback = 'Okay' THEN 3
            WHEN customer_feedback = 'Good' THEN 4
            WHEN customer_feedback = 'Very Good' THEN 5
        END AS feedback_score FROM order_t),
quarter_avg AS (SELECT quarter_number,
        AVG(feedback_score) AS avg_rating
    FROM feedback_mapped
    GROUP BY quarter_number),
overall_avg AS (SELECT AVG(feedback_score) AS overall_rating FROM feedback_mapped)
SELECT q.quarter_number, o.overall_rating, q.avg_rating
    FROM quarter_avg q
CROSS JOIN overall_avg o
ORDER BY q.quarter_number;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	overall_rating	avg_rating
1	3.1350	3.5548
2	3.1350	3.3550
3	3.1350	2.9563
4	3.1350	2.3970

Observations and Insights:

- Declining from Q1 – 3.55 to Q2 - 3.35 to Q3 – 2.96 to Q4 – 2.40. A sharp decline in the second half of the year suggests operational challenges Q2 to Q3 – 11.6% decrease in ratings and Q3 to Q4 18.9% decrease, which is a most drastic drop.
- Causes might be like low product quality, service delays, reduced incentives or pricing confusion
- To increase ratings should perform service quality improvements, loyalty programs, proactive engagement.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT quarter_number,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) / COUNT(*), 2)
AS very_bad_pct,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
bad_pct,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
okay_pct,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
good_pct,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) / COUNT(*), 2)
AS very_good_pct FROM order_t GROUP BY quarter_number ORDER BY quarter_number;
```

Output:

```
-- Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?
SELECT
    quarter_number,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) / COUNT(*), 2) AS very_bad_pct,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) / COUNT(*), 2) AS bad_pct,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) / COUNT(*), 2) AS okay_pct,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) / COUNT(*), 2) AS good_pct,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) / COUNT(*), 2) AS very_good_pct
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	very_bad_pct	bad_pct	okay_pct	good_pct	very_good_pct
1	10.97	11.29	19.03	28.71	30.00
2	14.89	14.12	20.23	22.14	28.63
3	17.90	22.71	21.83	20.96	16.59
4	30.65	29.15	20.10	10.05	10.05

Observations and Insights:

- Increasing customer dissatisfaction over time Q2: +3.92% from Q1, Q3 : +3.01% from Q2, Q4 +12.75% from Q3
- Decline in positive sentiment Q1 – 58.71% (highest customer satisfaction), Q2 – 50.77% (Moderate decline), Q3 – 37.55% (Noticeable drop), Q4 – 20.10% (Sharpest decline).
- Major satisfaction drop between Q3 and Q4, very bad ratings jumped by 12.75%, good ratings dropped by -17.45%.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```sql
SELECT quarter_number, COUNT(*) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

```sql
-- What is the trend of the number of orders by quarter?

SELECT quarter_number, COUNT(*) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	total_orders
1	310
2	262
3	229
4	199

Observations and Insights:

- Consistent decline in orders: Q1 - 310 orders (highest), Q2 – 262 orders (15.5% less from Q1), Q3 - 229 orders (12.6% less from Q2), Q4 – 199 orders (13.1% less from Q3) the steepest decrease in order volume.
- Decline customer retention: lower order volume across quarters hints at reduced repeat purchases or lost customers.
- Urgency for corrective measures: Stabilizing order volumes, Customer retention strategies, Operational improvement.

Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
WITH revenue_per_quarter AS (SELECT o.quarter_number,
        SUM(o.quantity * p.vehicle_price * (1 - o.discount)) AS net_revenue
    FROM order_t o
    JOIN product_t p ON o.product_id = p.product_id
    GROUP BY o.quarter_number),
revenue_with_lag AS (SELECT quarter_number, net_revenue,
        LAG(net_revenue) OVER (ORDER BY quarter_number) AS prev_revenue FROM revenue_per_quarter)
SELECT quarter_number, ceil(net_revenue) AS Net_revenue,
    round (CASE WHEN prev_revenue IS NULL THEN NULL
            ELSE ((net_revenue - prev_revenue) / prev_revenue) * 100
        END ) AS qoq_percentage_change
FROM revenue_with_lag ORDER BY quarter_number;
```

Output:

```
-- Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?
WITH revenue_per_quarter AS (
    SELECT o.quarter_number,
        SUM(o.quantity * p.vehicle_price * (1 - o.discount)) AS net_revenue
    FROM order_t o JOIN product_t p ON o.product_id = p.product_id
    GROUP BY o.quarter_number),
revenue_with_lag AS (
    SELECT quarter_number, net_revenue,
        LAG(net_revenue) OVER (ORDER BY quarter_number) AS prev_revenue
    FROM revenue_per_quarter)
SELECT quarter_number,
    ceil(net_revenue) AS Net_revenue,
    round (CASE
            WHEN prev_revenue IS NULL THEN NULL
            ELSE ((net_revenue - prev_revenue) / prev_revenue) * 100
        END ) AS qoq_percentage_change
FROM revenue_with_lag ORDER BY quarter_number;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	Net_revenue	qoq_percentage_change
1	18032550	NULL
2	13122996	-27
3	8882299	-32
4	8573150	-3

Observations and Insights:

- Net revenue shows a continuous decline across all quarters.
- The most drastic drop occurred between Q2 and Q3 (32% decline).
- Q4's revenue drop of 3% suggests a slight stabilization but still reflects weak performance.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```sql
SELECT
    o.quarter_number,
    round(SUM(o.quantity * (o.vehicle_price * (1 - o.discount)))) AS net_revenue,
    COUNT(o.order_id) AS total_orders
FROM order_t o
GROUP BY o.quarter_number
ORDER BY o.quarter_number;
```

Output:

```sql
-- What is the trend of net revenue and orders by quarters?
SELECT
    o.quarter_number,
    round(SUM(o.quantity * (o.vehicle_price * (1 - o.discount)))) AS net_revenue,
    COUNT(o.order_id) AS total_orders
FROM order_t o
GROUP BY o.quarter_number
ORDER BY o.quarter_number;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	net_revenue	total_orders
1	18032550	310
2	13122996	262
3	8882299	229
4	8573149	199

Observations and Insights:

- There is a steady decline in both net revenue (52% decline from Q1 to Q4) and total orders (36% decline from Q1 to Q4).
- The revenue drop is more pronounced than the order decline, indicating lower average revenue per order or increase discounting.
- Revenue per order drops by 26% from Q1 to Q4, possibly due to higher discounting, shifting product mix, or lower-margin sales.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```sql
SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount)*100, 2) AS avg_discount
FROM customer_t c
JOIN order_t o ON c.customer_id = o.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_discount DESC;
```

Output:

```sql
-- What is the average discount offered for different types of credit cards?
SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount)*100, 2) AS avg_discount
FROM customer_t c
JOIN order_t o ON c.customer_id = o.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_discount DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

credit_card_type	avg_discount
laser	64.38
mastercard	62.95
maestro	62.42
visa-electron	62.35
china-unionpay	62.22
instapayment	62.06
americanexpress	61.63
diners-club-us-ca	61.46
diners-club-carte-blanche	61.45
switch	61.02
bankcard	60.95
jcb	60.74
visa	60.08
diners-club-enroute	59.98
solo	58.50
diners-club-international	58.40

Observations and Insights:

- Laser cards get an average discount of 64.38%, the highest across all credit card types whereas Mastercard get an average of 62.95% the second highest across.
- High discounts on laser and Mastercard may drive higher customer loyalty and increased usage
- Recommendations for strategic action: review discount policy, targeted promotions, analyze customer behavior.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
    quarter_number,
    CEIL(AVG(DATEDIFF(ship_date, order_date))) AS avg_shipping_time
FROM order_t
WHERE ship_date IS NOT NULL AND order_date IS NOT NULL
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

```
-- What is the average time taken to ship the placed orders for each quarter?
SELECT
    quarter_number,
    CEIL(AVG(DATEDIFF(ship_date, order_date))) AS avg_shipping_time
FROM order_t
WHERE ship_date IS NOT NULL AND order_date IS NOT NULL
GROUP BY quarter_number
ORDER BY quarter_number;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	avg_shipping_time
1	58
2	72
3	118
4	175

Observations and Insights:

- Steady increase in average shipping time across quarters: Q1 – 58 days (fastest), Q2 – 72 days (+24.1%), Q3 118 days (+63.9%), Q4 - 175 days (+48.3%) the shipping duration has increased every quarter, with the largest jump between Q2 and Q3. This consistent upward trend could impact customer satisfaction and retention rates.
- Q4’s average shipping time (175 days) is 3x longer than Q1. This indicates operational inefficiencies, supply chain disruptions, or seasonal impacts.
- Recommendations to address rising shipping times: identify bottlenecks, optimize logistics, set realistic expectations, seasonal planning.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
48610993.76	1000	994	3.14
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
8573149.24	199	98	21.5

Business Recommendations

- Address decline revenue and order volume: optimize pricing and promotions, diversify product offerings, focus on high-performing regions.
- Improve operational efficiency: reduced shipping delays, monitor supply chain bottlenecks.
- Enhance customer experience: address declining satisfaction, personalized communications, flexible payment and credit options.
- Leverage data for strategic decisions: advanced analytics for demand forecasting, customer segmentation strategy, track key metrics in real time.
- Strengthen brand positioning: Capitalize on top vehicle makers, sustainability initiatives and customer advocacy programs.